

Card 1 — QUESTION 問題

Chimney

Card 2 — QUESTION 問題

Collar

Card 1 — ANSWER 答案

The tall vertical tube of the Bunsen burner — the flame comes out of the top.

Card 2 — ANSWER 答案

The ring around the lower part of the burner that you rotate to open or close the air hole.

Card 3 — QUESTION 問題

Air hole

Card 4 — QUESTION 問題

Luminous / quiet flame

Card 3 — ANSWER 答案

The opening on the collar. CLOSED → luminous flame. FULLY OPEN → roaring flame.

Card 4 — ANSWER 答案

Air hole CLOSED. Cooler flame, 900-950 °C.

Card 5 — QUESTION 問題

Non-luminous / roaring flame

Card 6 — QUESTION 問題

Heat-proof mat

Card 5 — ANSWER 答案

Air hole FULLY OPEN. Hotter flame, 1500-2000 °C.

Card 6 — ANSWER 答案

The mat the Bunsen burner stands on during heating.

Card 7 — QUESTION 問題

How to switch off a Bunsen burner

Card 8 — QUESTION 問題

Meniscus

Card 7 — ANSWER 答案

1. Close the air hole. 2. Turn off the gas tap. NEVER leave a flaming burner unattended.

Card 8 — ANSWER 答案

The curved surface of liquid in a cylinder. Read the volume at the **BOTTOM** of the meniscus, eyes level.

Card 9 — QUESTION 問題

Hypothesis

Card 10 — QUESTION 問題

Qualitative observation

Card 9 — ANSWER 答案

An educated guess — a possible explanation or answer to the question.

Card 10 — ANSWER 答案

Uses the senses (see, hear, feel, smell, taste). No numbers. E.g. “My finger is longer than yours.”

Card 11 — QUESTION 問題

Quantitative observation

Card 12 — QUESTION 問題

The 6 steps of scientific method

Card 11 — ANSWER 答案

Uses measurements and numbers. E.g. "My finger is 1 cm longer than yours."

Card 12 — ANSWER 答案

1 Question → 2 Research/Observation → 3 Hypothesis → 4 Experiment → 5 Analysis → 6 Conclusion.

Card 13 — QUESTION 問題

Fair test

Card 14 — QUESTION 問題

Independent variable

Card 13 — ANSWER 答案

An experiment that shows how ONE variable affects another variable.

Card 14 — ANSWER 答案

What you CHANGE in the experiment. (Plotted on the x-axis.)

Card 15 — QUESTION 問題

Dependent variable

Card 16 — QUESTION 問題

Controlled variable

Card 15 — ANSWER 答案

What you MEASURE in the experiment. (Plotted on the y-axis.)

Card 16 — ANSWER 答案

Kept CONSTANT so the test is fair. There can be more than one.

Card 17 — QUESTION 問題

Accuracy

Card 18 — QUESTION 問題

Precision

Card 17 — ANSWER 答案

How close a reading is to the TRUE value.

Card 18 — ANSWER 答案

How close repeated readings are to EACH OTHER.

Card 19 — QUESTION 問題

Reliability

Card 20 — QUESTION 問題

Line of best fit

Card 19 — ANSWER 答案

Precision + repeatability (same person & equipment) +
reproducibility (different person / equipment).

Card 20 — ANSWER 答案

The line closest to MOST points. Shows the trend and spots
anomalous data. NEVER dot-to-dot.

Card 21 — QUESTION 問題

Scientific diagram rules

Card 22 — QUESTION 問題

Hazard symbols learnt

Card 21 — ANSWER 答案

Pencil + ruler · at least 5 lines big · continuous strokes · label everything.

Card 22 — ANSWER 答案

Toxic · harmful · explosive · corrosive · highly flammable.